Scene-Adaptive Remote Sensing Image Super-Resolution Using a Multiscale Attention Network

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***Abstract*— Remote sensing image super-resolution has always been a major research focus, and many deep-learning-based algorithms have been proposed in recent years. However, since the structure of remote sensing images tends to be much more complex than that of natural images, several difficulties still remain for remote sensing images super-resolution. First, it is difficult to depict the nonlinear mapping between high-resolution (HR) and low-resolution (LR) images of different scenes with the same model. Second, the wide range of scales within the ground objects in remote sensing images makes it difficult for single-scale convolution to effectively extract features of various scales. To address the above-mentioned issues, we propose a multiscale attention network (MSAN) to extract the multilevel features of remote sensing images. The basic component of MSAN is the multiscale activation feature fusion block (MAFB). In addition, a scene-adaptive super-resolution strategy for remote sensing images is employed to more accurately describe the structural characteristics of different scenes. The experiments undertaken on several data sets confirm that the proposed algorithm outperforms the other state-of-the-art algorithms, in both evaluation indices and visual results.**

***Index Terms*— Channel attention, deep learning, multiscale activation, remote sensing imagery, scene adaptive.**

I. Introduction

SINGLE-IMAGE super-resolution (SISR) is a low-level computer vision problem referring to the technique of reconstructing a high-resolution (HR) image from a low-resolution (LR) image [1]–[3] to obtain more helpful detail information. On account of the wide application of SISR including satellite imaging, medical imaging, surveillance, and face recognition [4], it has been a research focus in the field of digital image processing for many years.

SISR is an inherently ill-posed problem because of the lack of redundant information in the LR images [5]–[7]. During the last few decades, many different frameworks have been developed to address this challenging problem. The existing SISR algorithms can be roughly classified into three categories [8]: 1) interpolation-based methods; 2) reconstruction-based methods; and 3) learning-based methods. Interpolation-based methods, including bilinear interpolation, bicubic interpolation [9], and the Lanczos resampling [10], are computationally efficient but usually lead to oversmoothing and the loss of high-frequency information. Reconstruction-based methods [11]–[14] usually utilize strong prior knowledge from the LR and HR image pairs to restrict the solution space of the reconstruction. These methods have an advantage in recovering sharp details, but they suffer from complex computation and performance degradation when the magnification factor increases. To alleviate the above-mentioned issues, learning-based methods have made great improvements in reconstruction accuracy, beginning with the Markov random field (MRF) model proposed by Freeman *et al.* [15]. Lately, random forest [16] and sparse coding methods [17]–[21] have been gradually introduced. Recently, due to the nonlinear representation and powerful feature extraction ability of deep neural networks, they have achieved great success in SISR. Many deep-learning-based algorithms have now been proposed for natural image super-resolution. These methods [22]–[24] usually utilize a convolutional neural network (CNN) to predict the nonlinear mapping relationship between the LR and HR patches from massive training data. Therefore, LR images can be reconstructed with an end-to-end framework, and these methods always outperform the previous traditional learning-based methods.

The above-mentioned deep-learning-based methods can achieve excellent reconstruction results on natural images. Compared with natural images, the structure of remote sensing images is more complex. First, remote sensing images usually have a much larger image width and contain a wide range of different scenes, such as buildings, farmland, forests, and airports. A complete remote sensing image can

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be composed of many different scenes, and the textural and structural information of them are quite different, resulting in inconsistency in the mapping relationship between the HR and LR images of different scenes. Second, the scales of the ground objects in remote sensing images vary greatly. For example, objects such as airplanes and vehicles occupy only a few pixels in remote sensing images, which is quite different from natural images. Consequently, it is difficult to accurately reconstruct the images of different scenes with the same super-resolution model, and the single-scale convolution cannot effectively extract features of different scales well. Many deep-learning-based frameworks for remote sensing image super-resolution have now been developed [25]–[27]. Although these methods can achieve a satisfactory result, to some extent, they tend to ignore the structural information of the remote sensing images [28], and they do not consider the underlying visual informativeness in different scenes. Therefore, in order to extract more useful knowledge from massive remote sensing images, a method that is more suitable for remote sensing images and can utilize the structural information between the different scenes in remote sensing images needs to be developed.

In this article, we propose a new framework that trains a deep neural network with remote sensing images from a different class of scenes to obtain different super-resolution models that are adapted to the remote sensing images of various scenes. A multiscale attention network (MSAN) is also presented to characterize the structural features of remote sensing images at multiple levels. In this article, we propose a multiscale activation module for the first time and then add a channel attention module behind. The contributions of our work can be concluded as threefold.

1) *Multiscale Activation Feature Fusion Block (MAFB):* We developed a new MAFB as the basic component of MSAN. Inspired by the wide activation strategy in [29], in this article, we first proposed a multiscale activation module to expand the number of features maps before the rectified linear unit (ReLU) activation layer and extract features at multilevels simultaneously in each MAFB. After fusing the feature maps of various scales, a channel attention module was then added behind to further adaptively utilize the effective information among the features of multiple scales.
2) *Scene-Adaptive Strategy:* We established a database of super-resolution models that were adapted to the different scenes of remote sensing images by transfer learning; 30 network models were trained in total for the remote sensing images of 30 different scenes contained in the Aerial Image data set (AID) 30] data set, which is commonly adopted for the task of aerial scene classification. Accordingly, based on this model database, a remote sensing image could be reconstructed scene-adaptively, after scene matching.
3) We analyzed the dissimilarities between the different scenes in remote sensing images through comparative experiments. In addition, the experimental results obtained on the AID data set and other remote sensing image data sets, including the Jilin-1 satellite images and the Massachusetts Road and Building Detection data set, all confirm the satisfactory performance of the proposed method.

The rest of this article is organized as follows. Section II introduces the existing SISR methods. The framework of the proposed MSAN and the details of the model are described in Section III. The simulated experimental results obtained with the AID data set, the Jilin-1 satellite images, and the Massachusetts Road and Building Detection data set, as well as a discussion, are given in Section IV. Finally, we present a conclusion of this work in Section V.

II. Related Work

This section presents a brief overview of the deep-learning-based SISR algorithms that are related to our work and introduces some of the existing super-resolution methods for remote sensing images.

A. SISR Based on Deep Learning

To address the SISR issue, the early deep-learning-based frameworks usually used interpolation techniques such as bicubic interpolation to super-resolve an LR image into HR space. Dong *et al.* [22] first introduced deep learning into the field of SISR by proposing a shallow super-resolution CNN (SRCNN) that contains three layers. Subsequently, inspired by the deep Res-Net network [31], Kim *et al.* [23] applied the concept of residual learning into super-resolution to speed up the convergence of the network and proposed two 20-layer networks termed very deep super-resolution (VDSR) and the deeply recursive convolutional network (DRCN) [24], which yielded better results than SRCNN. Lai *et al.* [32] proposed the deep Laplacian pyramid super-resolution network (LapSRN) to progressively predict the residual image from coarse to fine. In order to avoid the upsampling operation of LR images before the input into the network, Shi *et al.* [33] proposed an efficient subpixel convolutional network (ESPCN) by adding a pixel-shuffle layer at the tail of the network, making all the computation carried out in the LR space, which reduces the computational burden of the network. Later, an enhanced very deep and wide network known as the enhanced deep super-resolution network (EDSR) was proposed by Lim *et al.* [34], combining the advantages of residual learning and subpixel convolution and achieving excellent performance. Based on the SRResNet [35] model, EDSR optimizes the original residual block (RB) by removing the batch normalization layer to improve the computational efficiency and simplify the network, which has been proven to be an effective approach for the task of super-resolution. Yu *et al.* [29] improved the RBs of EDSR and proposed the WDSR model using the strategy of wide activation by expanding features before the ReLU activation. Zhang *et al.* [36] proposed a network named the very deep residual channel attention network (RCAN). They were the first to introduce the channel attention mechanism into image super-resolution, which is a mechanism that is more commonly used in high-level computer vision problems, including image classification and object recognition. The channel attention

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mechanism adaptively utilizes effective information among the feature maps by setting different scaling factors for the feature maps of different channels. Since then, more channel attention-based methods have been proposed [37]–[39]. Moreover, to fully exploit the extracted features, Li *et al.* [40] incorporated a multiscale module into the residual structure for the first time and achieved feature fusion at different scales.

B. SISR for Remote Sensing Images

Since remote sensing image super-resolution is of great significance for the subsequent object recognition [41], classification [42]–[44], and detection tasks [45], it has always been a major research focus. For remote sensing images, the enhancement of the spatial resolution is always dependent on the data fusion method [46], which is an effective approach, but has a bottleneck in the lack of simultaneous HR observations. To solve this limitation, SISR algorithms for remote sensing images have attracted increasing attention. The early methods [25]–[27] usually retrained a network designed for natural images, such as SRCNN and VDSR, with remote sensing images. The reconstruction results of these methods on remote sensing images are visually pleasing, to some extent, but the methods are less effective than when applied to natural images. Subsequently, considering the structural characteristics of remote sensing images, Ma *et al.* [47] decomposed the LR images into four subbands with a single-level 2-D discrete wavelet transform (WT) and then fed the four subbands into a recursive Res-Net to extract multiresolution features. Wang *et al.* [48] utilized 2-D-WT to describe the characteristics of the feature maps at different frequency levels via four SRCNN networks, which can greatly improve the representational power. Furthermore, in order to enhance the resolution of remote sensing images progressively, Jiang *et al.* [28] proposed a network named PECNN, which is composed of two subnetworks that extract low-level features and structural information, respectively. Lei *et al.* [49] proposed the local-global combined network (LGCNet) by leveraging the multilevel data representation power of deep learning for remote sensing images.

III. Methodology

As introduced in Section I, due to the complex structure of remote sensing images, there are two main difficulties in remote sensing images super-resolution. First, for the diverse scene objects in remote sensing images, the joint distribution pattern of the high-frequency information varies a lot, and thus, the high-and-low mapping relationship between different scenes tends to be inconsistent so that it is difficult to reconstruct remote sensing images of different scenes with the same model. Second, since the targets in remote sensing images often share a wide range of scales, the local texture information and global feature distribution of the image should be taken into account. Thus, compared with natural images, it is of great significance to learn multilevel features and enforce feature expression. To address the aforementioned issues, based on the WDSR architecture, we first optimize it by proposing a new multiscale activation feature fusion RB, which is called MAFB. In each MAFB, we combine a multiscale activation module and a channel attention module together. Then, a scene-adaptive framework for remote sensing image super-resolution is developed. In this part, the proposed MSAN model and the scene-adaptive strategy are described in detail.

A. MSAN for Remote Sensing Image Super-Resolution

The architecture of the proposed MSAN model is demonstrated in Fig. 1, including the complete network structure and the new RB MAFB, as shown in Fig. 1(a) and (b), respectively. The overall structure of MSAN consists of two components: 1) the identity mapping and 2) the residual body. For the identity mapping branch, we utilize the original 5×5 convolutional layer to extract LR features and keep the upsampling layer with the subpixel convolution of WDSR. In this article, we aim at recovering a super-resolution remote sensing image from the LR input, as follows:

$$I^{\mathrm{SR}} = F(I^{\mathrm{LR}}) \tag{1}$$

where F is the mapping learned by the end-to-end network, $I^{\mathrm{LR}} \in R^{H\times W\times c}$ stands for the LR input, $I^{\mathrm{SR}} \in R^{sH\times sW\times c}$ represents the SR output, and s is the scale factor. Assuming that there are N MAFBs in the residual branch, the input of the first and the ith ($i \in [1, N)$) blocks can be defined as

$$R_0 = \omega_{3\times 3} * I^{\mathrm{LR}} + b_0 \tag{2}$$

$$R_i = \delta_i(R_{i-1}) = \delta_i(\delta_{i-1}(\ldots\delta_1(R_0)\ldots)) \tag{3}$$

where ω and b represent the filter and bias, respectively, the subscripts represent the kernel size, and δ represents the function of each MAFB. The size of $\omega_{3\times 3}$ is $c \times 3 \times 3 \times n$, where $c = 3$ stands for the RGB channels of the input LR image and n is the number of output feature maps.

B. Multiscale Activation Feature Fusion Block (MAFB)

The MAFB is developed upon a multiscale activation module, a feature fusion bottleneck layer, and a channel attention module, as shown in Fig. 1(b).

1) Multiscale Activation Module: Since it has been proven by WDSR that expanding the feature maps before the ReLU activation contributes to the performance of the network, thus, in this article, we first propose the multiscale activation module. We extend the wide activation strategy to multiscale activation by adding a multiscale convolution module before the ReLU activation in each MAFB to expand the number of features maps of various scales and extract the interclass diversities of the different scenes at multiple scales simultaneously, as shown in Fig. 1(b). The kernel sizes of the multiscale convolutional filters are 3×3, 5×5, and 7×7, respectively. The multiscale convolution module has the ability to describe the contextual and textural information of the input LR image at different levels and can enhance the representational capacity of the network. Since the multiscale convolution operation has receptive fields of different sizes, long-term dependencies

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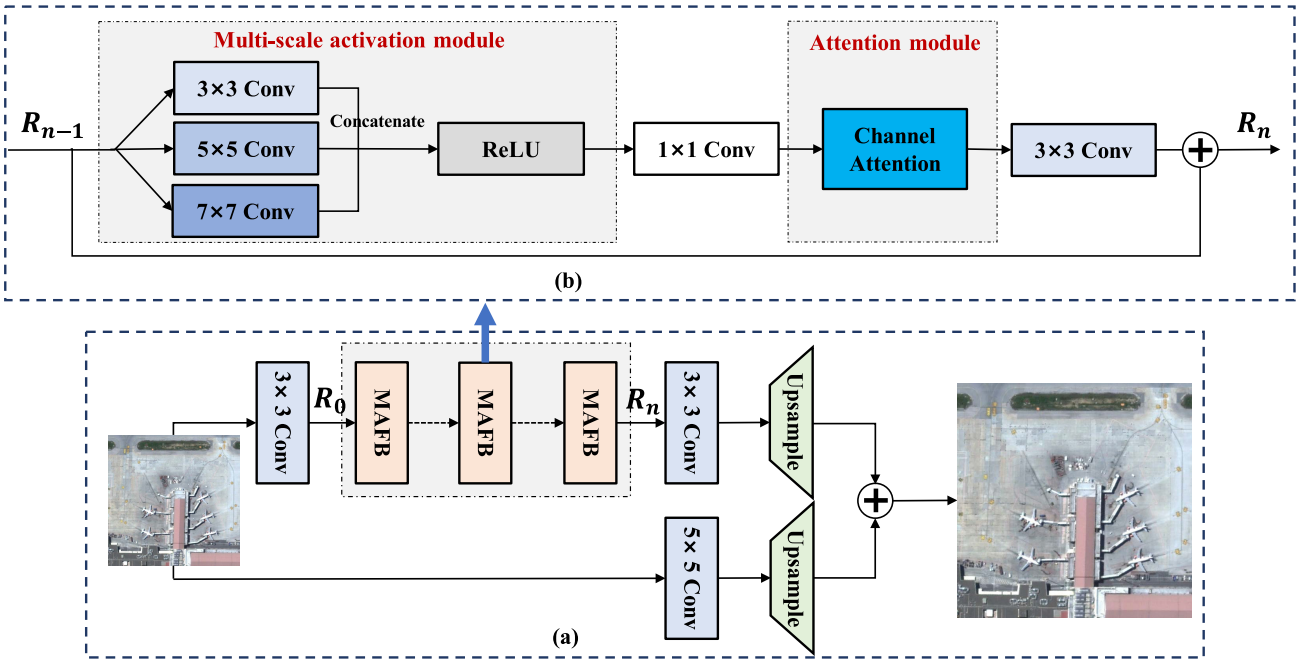


Fig. 1. Proposed MSAN model. (a) Complete network structure. (b) Multiscale activation feature fusion block (MAFB).

can be processed after passing through several RBs. Moreover, with the combination of three kernel sizes, multilevel features, and larger receptive fields can be obtained so that the joint information of the local and global distribution of different scenes in the remote sensing images can be captured at the same time. The multiscale convolution module in our MAFB can be formulated as

$$C_1 = \omega_{3\times3} * R_{n-1} + b_1 \tag{4}$$

$$C_2 = \omega_{5\times5} * R_{n-1} + b_2 \tag{5}$$

$$C_3 = \omega_{7\times7} * R_{n-1} + b_3. \tag{6}$$

We concatenate the output feature maps of the three convolutions together and then apply the ReLU function

$$M = \sigma[\text{concat}(C_1, C_2, C_3)] \tag{7}$$

where $\sigma(\cdot)$ denotes the nonlinear activation function and $M \in R^{H\times W\times 3n}$. If we suppose that each of the three convolution operations outputs n feature maps, then we can obtain $3\times n$ feature maps after the ReLU activation. Subsequently, a 1×1 convolution is followed to fuse the multiscale features extracted before. Also, it still acts as a bottleneck layer to reduce the channel number to n

$$M' = \omega_{1\times1} \times M + b' \tag{8}$$

where $M' \in R^{H\times W\times n}$.

2) Attention Module: The channel attention module in the RCAN architecture is shown in Fig. 2. The channel attention mechanism was first introduced into the field of super-resolution by RCAN, and it has been proven to have the ability to adaptively utilize effective information among the different channels. In terms of remote sensing images, the high-frequency components in LR images are valuable, especially for the scenes that are full of edges, texture, and

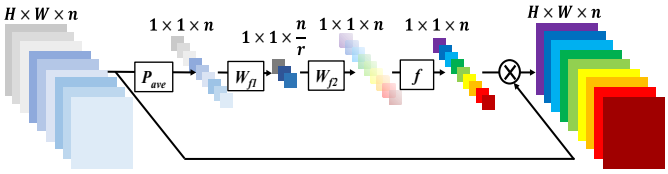


Fig. 2. Channel attention module in RCAN. $\otimes$ denotes the elementwise product.

other details. In order to enforce the feature expression and make the network focus on more informative features, we add the attention module shown in Fig. 2 into the RBs to rescale the channelwise features after the 1×1 convolution, which can be performed as follows:

$$S = f(W_{f_2} * \sigma(W_{f_1} * P_{\text{ave}}(M'))) \tag{9}$$

where $P_{\text{ave}}(M') \in R^{1\times1\times n}$ represents the global average pooling operation, $W_{f1} \in R^{1\times1\times(n/r)}$ and $W_{f2} \in R^{1\times1\times n}$ represent two fully connected layers, and $f(\cdot)$ denotes the sigmoid function. After the recalibration operation, we use a 3×3 convolution for spatialwise feature extraction at the tail of the RB.

Considering the parameter budget and computational burden of the network, the number of feature maps n is set to 64. For the sake of experimental efficiency, we use eight RBs as the baseline in the proposed MSAN and two comparative approaches (i.e., EDSR and WDSR). Moreover, weight normalization (WN) is utilized to improve the efficiency and accuracy of the training. The l_1 loss function between the ground-truth and the super-resolution images is also calculated to train the final MSAN.

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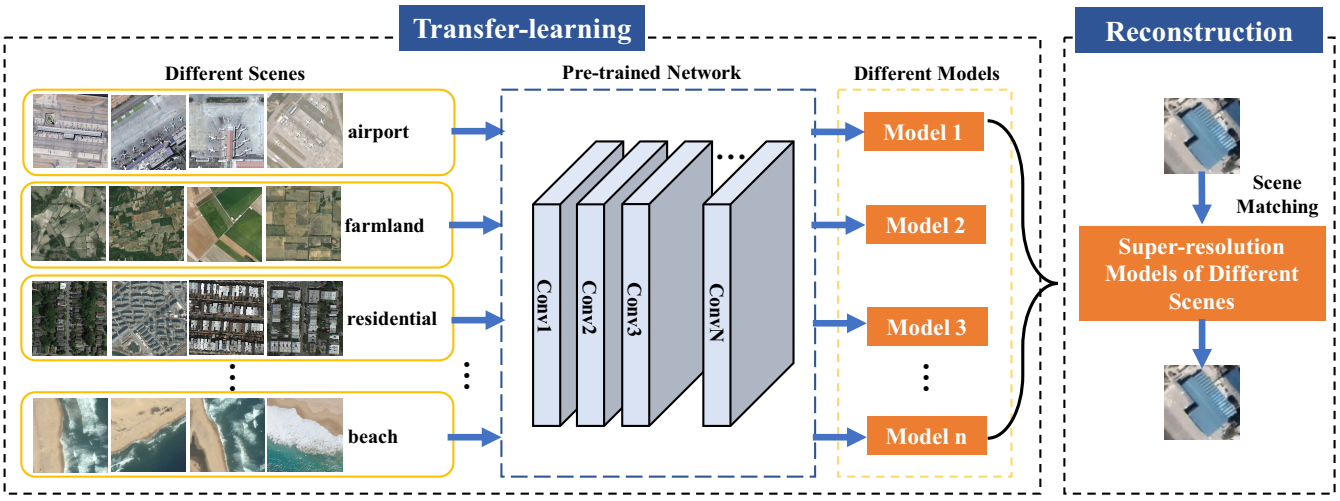


Fig. 3. Flowchart of the proposed method of scene-adaptive super-resolution.

TABLE I

MODEL COMPARISONS AT DIFFERENT NUMBERS OF RBS TRAINED ON AID DATA SET. THE FIGURES ARE THE AVERAGE PSNR/SSIM VALUES TESTED ON 900 VALIDATION IMAGES FROM 30 CLASSES OF SCENES. THE FIGURES IN BOLD REPRESENT THE BEST PERFORMANCE

×2	Number of RBs	1			3		
	Networks	EDSR	WDSR	Proposed	EDSR	WDSR	Proposed
	PSNR/SSIM	36.089/0.9357	36.159/0.9363	**36.315/0.9374**	36.308/0.9389	36.381/0.9395	**36.483/0.9400**
	Number of RBs	5			8		
	Networks	EDSR	WDSR	Proposed	EDSR	WDSR	Proposed
	PSNR/SSIM	36.550/0.9406	36.601/0.9410	**36.716/0.9417**	36.676/0.9408	36.715/0.9411	**36.828/0.9420**
×4	Number of RBs	1			3		
	Networks	EDSR	WDSR	Proposed	EDSR	WDSR	Proposed
	PSNR/SSIM	29.878/0.7772	29.940/0.7785	**30.103/0.7820**	30.056/0.7816	30.136/0.7823	**30.244/0.7885**
	Number of RBs	5			8		
	Networks	EDSR	WDSR	Proposed	EDSR	WDSR	Proposed
	PSNR/SSIM	30.278/0.7940	30.359/0.7953	**30.470/0.7975**	30.368/0.7960	30.469/0.7971	**30.578/0.7991**

C. Scene-Adaptive Framework Description

Considering the great difference between the different scenes in remote sensing images, it is difficult to train a model that works well for all the diverse scenes. Therefore, in this article, we propose a scene-adaptive super-resolution framework. The flowchart of the proposed framework is shown in Fig. 3. As shown in Fig. 3, the scene-adaptive super-resolution flowchart can be divided into two parts: 1) transfer learning and 2) reconstruction.

1) Transfer Learning: First, we pretrain a basic network with remote sensing images containing all kinds of scenes to initialize the parameters. However, due to the conspicuous disparity of the texture information between the different scenes and the wide range of scales among the remote sensing images, the initialized model tends to be a compromised result of the game between the training data of different scenes. Thus, this pretrained model will not achieve the best reconstruction result for each single scene. For this reason, to allow the initialized parameters to be transferred to the domain of the corresponding scene of the remote sensing images, we use remote sensing images of different scenes to fine-tune the pretrained basic network [50].



Fig. 4. Examples of the 30 categories of different scenes in the AID data set.

After obtaining the pretrained basic network, the remote sensing images of different scenes are used to fine-tune the pretrained network to obtain the final models. Therefore, we can establish a database that contains super-resolution models adapted to the different scenes of remote sensing images. Meanwhile, we can utilize the structural information of the different scenes to explore a more accurate nonlinear mapping between the LR and HR images to improve the reconstruction results.

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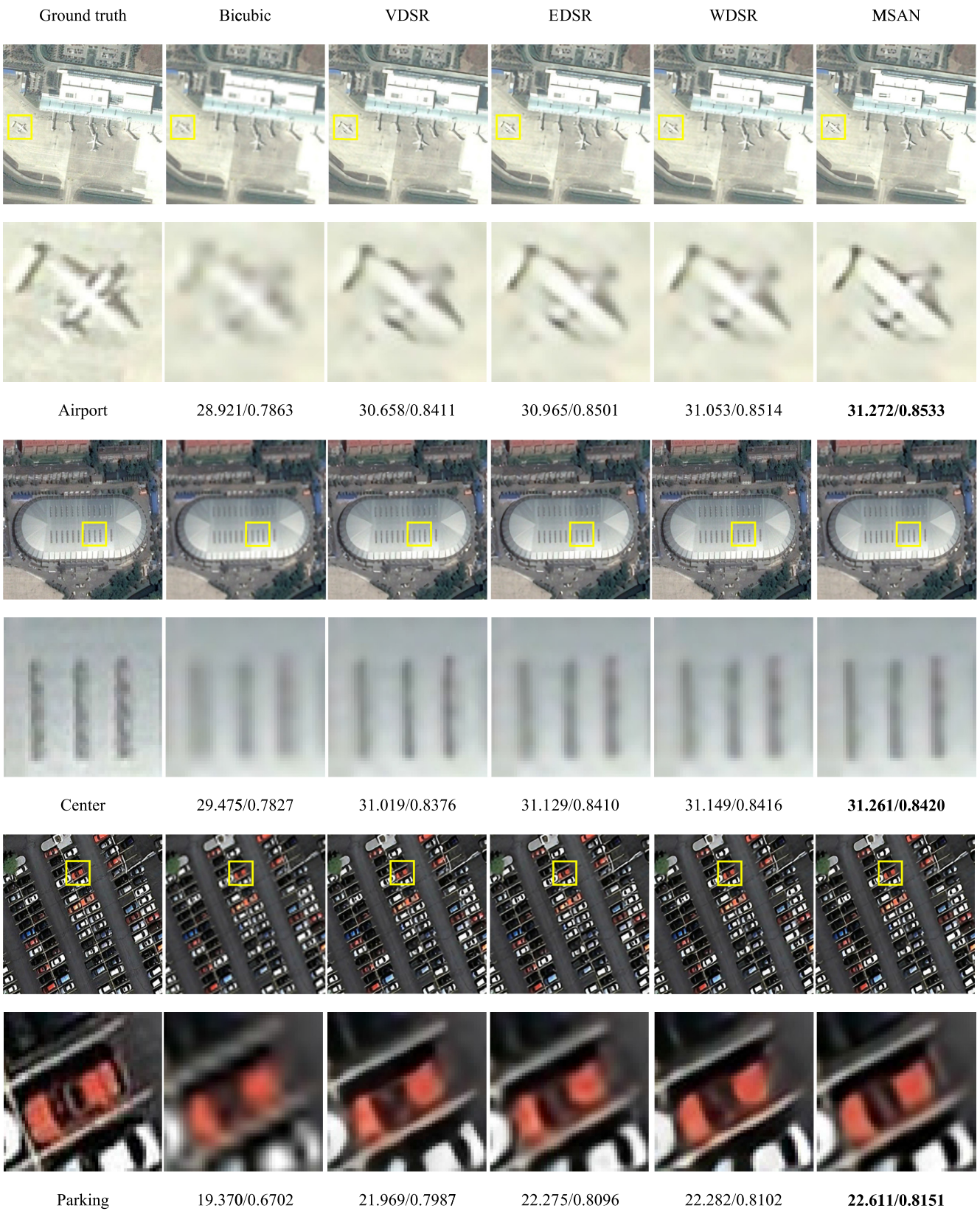


Fig. 5. Visual comparisons and PSNR/SSIM value of the proposed method and other methods on the scene of the airport, center, and parking for scale ×4. The figures in bold represent the best performance.

2) Reconstruction: In the reconstructed part, after obtaining all the models fine-tuned with the different scenes, we can apply the scene-adaptive super-resolution framework. First, we can match a remote sensing image to the corresponding category with scene classification. After scene matching, each remote sensing image is input into the corresponding

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TABLE II
RUN-TIME COMPARISON FOR 100 TESTING IMAGES

Networks	EDSR	WDSR	MASN
Seconds	2.04s	1.68s	3.20s

super-resolution model that was obtained in the first procedure adaptively to obtain the final HR output.

IV. EXPERIMENTS AND DISCUSSION

A. Data Set and Experimental Settings

The AID data set is a large-scale benchmark data set that is commonly adopted for the task of scene classification of remote sensing images. This data set contains 30 classes of different scene categories, as shown in Fig. 4. Specifically, the 30 classes are airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct. The numbers of sample images vary a lot with different aerial scene types, from 220 up to 420. The AID data set contains more than 10 000 images with multiple-resolutions ranging from 8 to 0.5 m. The size of each aerial image was fixed as 600 × 600 pixels, which can cover scenes with various scales.

First, we selected 900 images from the AID data set, made up of 30 images for each category, as the pretraining data set to pretrain a basic MSAN with the ADAM optimizer for 300 epochs. In addition, a total of 30 images, made up of one image for each category, were randomly selected as the testing data set We set $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$. All the LR images were cropped to 48 × 48 as inputs, and the batch size was set to 16 when pretraining. The initial learning rate was set as 1×10^{-3} with WN and halved every 100 epochs. After obtaining the pretrained network, we randomly selected 150 images from each of the 30 classes of different scenes as the training data set and ten for the testing to fine-tune the pretrained network, and the learning rate is fixed to 1×10^{-4} when fine-tuning the 30 models. In addition, another 30 images were selected from each category for the validation. It is important to note that all the training was carried out on the RGB channels. During the pretraining and fine-tuning, we augmented the training data by randomly flipping the images horizontally and rotating them 90° and then subtracted the average RGB values of the corresponding training data set from each LR image.

The pretraining and fine-tuning operations were carried out under the PyTorch framework. All 30 models were fine-tuned over 15 epochs. It took roughly 12 h to pretrain the MSAN, and each model of the different scenes took about an hour to fine-tune using NVIDIA GTX2080 GPUs. Eventually, we, respectively, obtained 30 models for all the scene categories in the AID data set based on MSAN.

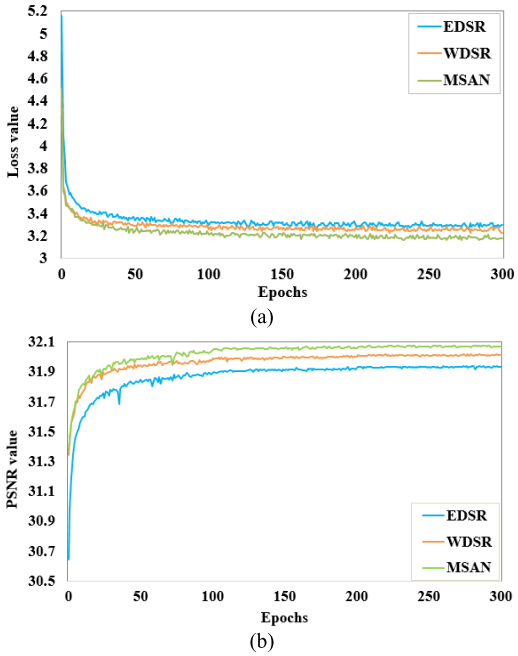


Fig. 6. (a) Training l_1 loss of EDSR, WDSR, and MSAN. (b) Validation PSNR values of EDSR, WDSR, and MSAN during training.

B. Comparison With Other CNN-Based Methods

1) Experiments on RBs: To compare the reconstruction accuracy at different model depths, a set of experiments is carried out for both 2× and 4× SR on the AID data set. For the sake of fairness, we tested the proposed MSAN model and the baseline EDSR and WDSR with a different number of RBs that were trained on the same pretraining data set. The average quantitative evaluation results calculated on the test images from 30 categories of scenes are shown in Table I. It can be seen that the proposed MSAN model achieves the highest peak-signal-to-noise ratio (PSNR) and structural similarity (SSIM) values on the proposed MAFB.

2) Ablation Study: Moreover, we carried out an ablation experiment to test the performance of the multiscale activation and channel attention modules in the proposed MFAB. The ablation study was conducted on the final model that contains eight RBs. The results given in Table II confirm that the multiscale activation module contributes more to the improvement than the channel attention model. The multiscale activation module can improve the performance of the network by 0.182 and 0.176 dB for the results of 2× and 4× SR, respectively, while the channel attention module can improve by 0.072 and 0.062 dB. Obviously, the model with both the multiscale activation and channel attention module can achieve the best performance.

3) Reconstruction Results Comparison: Meanwhile, we evaluated the performance of the pretrained MSAN model on the tested images through all the categories of scenes and compared the results with those of bicubic, VDSR, EDSR, and WDSR methods. For a fair and convincing comparison, we retrained these networks with the same training data set as the pretrained MSAN to obtain their best

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TABLE III

ABLATION STUDIES WITH DIFFERENT COMPONENTS OF THE PROPOSED RB FOR DIFFERENT SCALES

	Multi-scale	Channel attention	PSNR/SSIM		Multi-scale	Channel attention	PSNR/SSIM
	×	×	36.600/0.9396		×	×	30.363/0.7955
×2	×	√	36.672/0.9399	**×4**	×	√	30.425/0.7962
	√	×	36.782/0.9412		√	×	30.539/0.7982
	√	√	**36.828/0.9420**		√	√	**30.578/0.7991**

TABLE IV

QUANTITATIVE EVALUATION RESULTS OF BICUBIC, VDSR, EDSR, WDSR, AND THE PROPOSED MSAN TRAINED WITH THE AID DATA SET. THE FIGURES PRESENT THE AVERAGE PSNR/SSIM ON 30 IMAGES OF EACH CLASS OF SCENE FOR SCALE ×2. THE FIGURES IN BOLD PRESENT THE BEST PERFORMANCE

scale	Classes	Bicubic	VDSR	EDSR	WDSR	MSAN
	Airport	35.797 / 0.9316	38.391 / 0.9584	38.547 / 0.9611	38.641 / 0.9617	**38.822 / 0.9619**
	Bare land	43.949 / 0.9763	45.679 / 0.9846	45.999 / 0.9854	46.003 / 0.9854	**46.038 / 0.9856**
	Baseball field	33.444 / 0.8909	35.855 / 0.9285	36.130 / 0.9301	36.175 / 0.9305	**36.353 / 0.9315**
	Beach	30.265 / 0.8510	32.014 / 0.8994	32.125 / 0.9008	32.141 / 0.9010	**32.176 / 0.9015**
	Bridge	38.810 / 0.9556	41.294 / 0.9723	41.865 / 0.9737	41.905 / 0.9739	**42.021 / 0.9744**
	Center	31.169 / 0.8741	33.799 / 0.9243	34.162 / 0.9272	34.218 / 0.9279	**34.389 / 0.9298**
	Church	28.237 / 0.8395	30.765 / 0.9008	30.958 / 0.9031	31.022 / 0.9037	**31.159 / 0.9056**
	Commercial	33.889 / 0.9257	36.668 / 0.9577	37.023 / 0.9594	37.040 / 0.9596	**37.167 / 0.9604**
	Dense-residential	28.353 / 0.8251	30.656 / 0.8923	30.827 / 0.8947	30.858 / 0.8951	**30.935 / 0.8969**
	Desert	44.631 / 0.9693	46.032 / 0.9774	46.142 / 0.9778	46.164 / 0.9779	**46.196 / 0.9780**
	Farmland	39.400 / 0.9444	41.486 / 0.9627	41.730 / 0.9640	41.756 / 0.9642	**41.871 / 0.9651**
	Forest	34.202 / 0.8667	35.900 / 0.9122	36.002 / 0.9134	36.010 / 0.9134	**36.081 / 0.9148**
	Industrial	31.375 / 0.8906	34.468 / 0.9402	34.750 / 0.9423	34.813 / 0.9429	**34.926 / 0.9441**
	Meadow	40.832 / 0.9356	42.610 / 0.9559	42.805 / 0.9567	42.817 / 0.9567	**42.892 / 0.9573**
	Medium-residential	30.967 / 0.8728	33.678 / 0.9222	33.897 / 0.9240	33.965 / 0.9244	**34.122 / 0.9258**
×2	Mountain	36.840 / 0.9316	38.951 / 0.9580	39.079 / 0.9588	39.090 / 0.9589	**39.121 / 0.9592**
	Park	32.159 / 0.8961	34.556 / 0.9375	34.771 / 0.9390	34.797 / 0.9393	**34.875 / 0.9401**
	Parking	31.924 / 0.9319	34.877 / 0.9620	35.647 / 0.9650	35.704 / 0.9654	**36.019 / 0.9668**
	Playground	31.365 / 0.8762	34.522 / 0.9262	34.889 / 0.9292	34.980 / 0.9300	**35.177 / 0.9316**
	Pond	34.200 / 0.9043	35.977 / 0.9317	36.120 / 0.9327	36.147 / 0.9329	**36.212 / 0.9334**
	Port	31.495 / 0.9313	34.302 / 0.9590	34.659 / 0.9606	34.701 / 0.9608	**34.836 / 0.9614**
	Railway Station	33.252 / 0.8990	35.850 / 0.9409	36.070 / 0.9426	36.111 / 0.9430	**36.228 / 0.9442**
	Resort	37.652 / 0.9517	40.184 / 0.9719	40.619 / 0.9732	40.646 / 0.9734	**40.739 / 0.9739**
	River	38.063 / 0.9265	39.734 / 0.9492	39.847 / 0.9498	39.858 / 0.9499	**39.897 / 0.9504**
	School	30.326 / 0.8639	32.744 / 0.9144	32.919 / 0.9163	32.965 / 0.9169	**33.037 / 0.9180**
	Sparse-residential	30.896 / 0.8342	32.612 / 0.8912	32.735 / 0.8931	32.757 / 0.8933	**32.866 / 0.8952**
	Square	31.895 / 0.9170	35.202 / 0.9573	35.677 / 0.9594	35.731 / 0.9598	**35.933 / 0.9607**
	Stadium	32.323 / 0.9100	34.927 / 0.9433	35.283 / 0.9456	35.345 / 0.9461	**35.469 / 0.9470**
	Storage Tanks	29.144 / 0.8446	31.376 / 0.8983	31.519 / 0.9003	31.574 / 0.9010	**31.635 / 0.9020**
	Viaduct	34.671 / 0.9047	37.236 / 0.9412	37.472 / 0.9428	37.525 / 0.9433	**37.637 / 0.9443**
	Average	34.051 / 0.9024	36.411 / 0.9393	36.676 / 0.9408	36.715 / 0.9411	**36.828 / 0.9420**

performance. Table III lists the final mean PSNR and SSIM values over the test images of all 30 categories of scenes under these methods. The proposed MSAN model achieves the highest PSNR and SSIM values for all 30 classes of scenes. Moreover, the average PSNR over the 30 scenes with MSAN is 0.152 and 0.113 dB higher than that of the EDSR and WDSR methods, respectively. However, as revealed in Table III, the improvement degree of the pretrained MSAN model in the reconstruction results varies a lot between different scenes. For instance, for scenes such as parking and square, the PSNR value increases by 0.315 and 0.202 dB, respectively, compared with the results of WDSR, while for other scenes such as desert and river, the PSNR values increase only slightly. This encourages us to further explore the impact of scene differences on the reconstruction results in the following.

Fig. 5 shows some of the visual results of the different methods. It is clear that the reconstructed images of MSAN are much better than those of the other methods. By comparing the enlarged parts, MSAN shows a better SR performance in preserving the texture information and edges. For example, in Fig. 5, with MSAN, the outline of the airplane, such as the wings and the tail, is clearly enhanced. In addition, in the parking scene image shown in Fig. 5, the reconstructed red cars obtained by the comparative methods are blurred and distorted to some extent, while the car obtained by MSAN is

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TABLE V

QUANTITATIVE EVALUATION RESULTS OF THE PRETRAINED MSAN AND FINE-TUNED SAMSAN APPLYING SCENE THE ADAPTIVE-STRATEGY. THE FIGURES ARE THE AVERAGE PSNR/SSIM VALUES ON 30 TEST IMAGES OF EACH CLASS OF SCENE FOR SCALES ×2 AND ×4. THE FIGURES IN BOLD REPRESENT THE BEST PERFORMANCE

Classes	scale	MSAN	SAMSAN	scale	MSAN	SAMSAN
Airport	×2	38.822 / 0.9619	**39.047 /0.9622**	×4	32.587 / 0.8561	**32.755 / 0.8596**
Bare land		46.038 / 0.9856	**46.154 / 0.9856**		38.701 / 0.9235	**38.832 / 0.9247**
Baseball field		36.353 / 0.9315	**36.544 / 0.9318**		30.614 / 0.8001	**30.759 / 0.8037**
Beach		32.176 / 0.9015	**32.291 / 0.9020**		27.045 / 0.7150	**27.218 / 0.7194**
Bridge		42.021 / 0.9744	**42.303 / 0.9750**		34.245 / 0.8908	**34.562 / 0.8936**
Center		34.389 / 0.9298	**34.581 / 0.9321**		28.175 / 0.7762	**28.391 / 0.7835**
Church		31.159 / 0.9056	**31.487 / 0.9099**		25.397 / 0.7024	**25.806 / 0.7184**
Commercial		37.167 / 0.9604	**37.457 / 0.9617**		29.594 / 0.8188	**29.914 / 0.8261**
Dense-residential		30.935 / 0.8969	**31.196 / 0.9004**		25.393 / 0.6780	**25.686 / 0.6971**
Desert		46.196 / 0.9780	**46.228 / 0.9782**		41.206 / 0.9361	**41.283 / 0.9365**
Farmland		41.871 / 0.9651	**42.148 / 0.9666**		35.984 / 0.8784	**36.125 / 0.8810**
Forest		36.081 / 0.9148	**36.162 / 0.9163**		30.200 / 0.6856	**30.315 / 0.6940**
Industrial		34.926 / 0.9441	**35.169 / 0.9453**		28.272 / 0.7718	**28.531 / 0.7801**
Meadow		42.892 / 0.9573	**42.977 / 0.9581**		36.535 / 0.8462	**36.576 / 0.8470**
Medium-residential		34.122 / 0.9258	**34.446 / 0.9276**		27.740 / 0.7275	**28.104 / 0.7381**
Mountain		39.121 / 0.9592	**39.255 / 0.9603**		32.692 / 0.8240	**32.886 / 0.8295**
Park		34.875 / 0.9401	**35.029 / 0.9418**		28.217 / 0.7617	**28.439 / 0.7683**
Parking		36.019 / 0.9668	**36.492 / 0.9679**		27.113 / 0.8235	**27.656 / 0.8340**
Playground		35.177 / 0.9316	**35.502 / 0.9333**		28.771 / 0.7800	**29.134 / 0.7895**
Pond		36.212 / 0.9334	**36.306 / 0.9342**		31.689 / 0.8304	**31.798 / 0.8322**
Port		34.836 / 0.9614	**35.212 / 0.9628**		27.770 / 0.8554	**28.233 / 0.8620**
Railway Station		36.228 / 0.9442	**36.475 / 0.9459**		29.777 / 0.7835	**30.012 / 0.7911**
Resort		40.739 / 0.9739	**40.834 / 0.9741**		32.838 / 0.8648	**32.957 / 0.8675**
River		39.897 / 0.9504	**39.955 / 0.9510**		34.634 / 0.8378	**34.716 / 0.8392**
School		33.037 / 0.9180	**33.243 / 0.9184**		27.766 / 0.7576	**27.931 / 0.7640**
Sparse-residential		32.866 / 0.8952	**33.006 / 0.8980**		27.366 / 0.6304	**27.522 / 0.6400**
Square		35.933 / 0.9607	**36.179 / 0.9619**		28.504 / 0.8230	**28.774 / 0.8289**
Stadium		35.469 / 0.9470	**35.731 / 0.9474**		29.733 / 0.8378	**30.031 / 0.8444**
Storage Tanks		31.635 / 0.9020	**31.808 / 0.9043**		27.493 / 0.7477	**27.636 / 0.7528**
Viaduct		37.637 / 0.9443	**37.772 / 0.9452**		31.294 / 0.8083	**31.505 / 0.8139**
Average		36.828 / 0.9420	**37.033 / 0.9433**		30.578 / 0.7991	**30.803 / 0.8053**

much sharper, and the PSNR value is increased by 0.329 dB compared with WDSR.

4) Training and Run-Time Comparison: To further compare the convergence and accuracy when training EDSR, WDSR, and MSAN, the loss and PSNR value curves of the three networks are plotted in Fig. 6, respectively. In addition, the average running times of the three models are also recorded in Table IV to compare the work efficiency. Referring to Table IV, it can be seen that the time taken by MSAN for super-resolving 100 LR images is almost close to the other two algorithms, but MSAN has both faster convergence and better accuracy when training, as shown in Fig. 6.

C. Results of the Scene-Adaptive Super-Resolution Strategy

According to the preliminary experimental results given in Section B, our proposed MAFB works well for remote sensing images and the pretrained MSAN model already outperforms the comparative algorithms. In order to further improve the reconstruction results of different land-cover scenes exiting in remote sensing images, the scene-adaptive strategy has been applied. First, we established the database of super-resolution models that adapted to the different scenes of remote sensing images by transfer learning. Images from different scene categories were utilized to fine-tune the pretrained MSAN. Second, comparison experiments on all 30 scene categories of AID data set with the pretrained basic network and the fine-tuned model were conducted. For convenience, MSAN refers to the pretrained model, and the fine-tuned model with the scene-adaptive strategy is denoted as SAMSAN. We tested each model on the corresponding scene. The quantitative evaluation SAMSAN and MSAN are given in Table V, and the five scenes for which the average PSNR value of the test images increases the results for the test images are listed in Table V. It is worth noting that the scene-adaptive strategy yields a great improvement in terms of the index values of PSNR and SSIM on all 30 classes of scenes. The average PSNR value on the test images of all 30 scenes increases by 0.205 and 0.225 dB for scales ×2 and ×4, respectively, after the fine-tuning operation (SAMSAN) compared with the pretrained model (MSAN). Although the pretrained MSAN model is already able to achieve better results than WDSR and EDSR, it is found that the super-resolution results are further improved after applying the scene-adaptive strategy.

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Fig. 7. Visual comparisons and PSNR/SSIM values of the pretrained MSAN and SAMSAN on the scene of the church, port, medium residential, and commercial for scale $\times 4$. The figures in bold represent the best performance.

In fact, images of different scenes have different structural characteristics. Scenes such as bare land, desert, and forest tend to be smoother and, therefore, contain less structural information. Conversely, other scenes such as airport, parking, and school contain more high-frequency details such as edges and texture information, and thus, much redundant information

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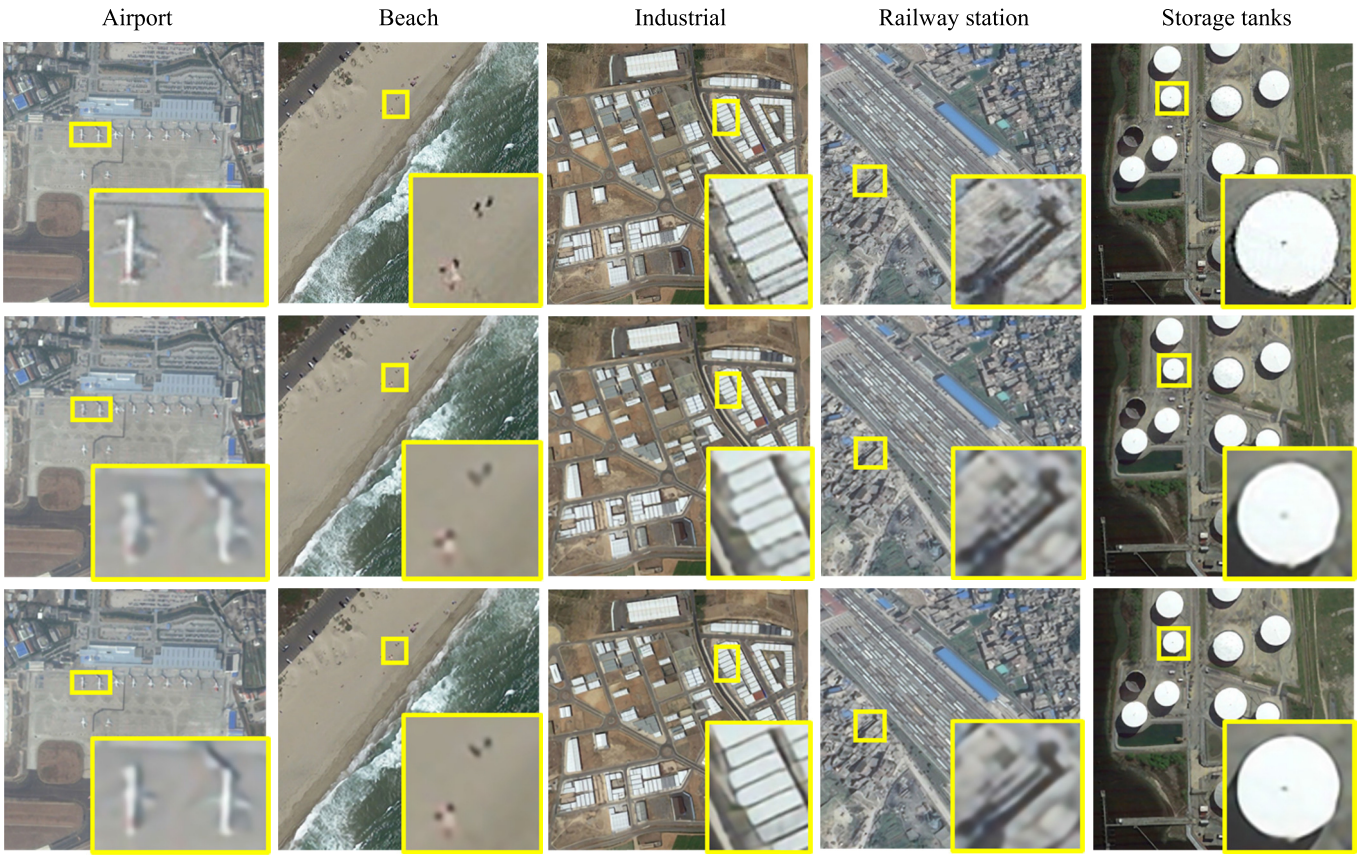


Fig. 8. Visual comparisons of the reconstruction results obtained using the model fine-tuned on the forest scenes to perform inference on the scenes of airport, beach, industrial, railway station, and storage tanks for scale $\times 4$. The first line is the ground-truth images. The second line is the reconstruction results obtained using the model fine-tuned on the forest scenes. The last line is the result reconstructed by their own model.

can be utilized during the fine-tuning. Generally speaking, comparing the results of SAMSAN and MSAN, the five scenes of which the average PSNR value increase the most are parking, port, church, medium-residential, and playground. The PSNR values of these five scenes increase by 0.543, 0.463, 0.409, 0.364, and 0.363 dB, respectively, for scale $\times 4$ after the fine-tuning operation. At the same time, the five scenes for which the PSNR value increases the least are meadow, desert, river, pond, and forest. The PSNR values for these five scenes increase by 0.041, 0.076, 0.081, 0.109, and 0.114 dB, respectively, for scale $\times 4$ after the fine-tuning. The results for scales $\times 2$ and $\times 4$ are almost consistent. It is clear from the above-mentioned results that the scene-adaptive strategy works better for those scenes containing richer texture information.

A visual comparison of the results for several typical scenes is shown in Fig. 7. It is clear that for the church scene, SAMSAN generates more convincing results with fewer artifacts than the pretrained MSAN. Furthermore, the scene-adaptive strategy helps to enhance the edges of the objects in remote sensing images, as in the reconstruction results of SAMSAN for the port and medium-residential scenes presented in Fig. 7. In addition, in terms of the commercial scene, the reconstruction result obtained by SAMSAN is less distorted and contains more recovered details.

In order to further explore the impact of the scene differences on the reconstruction results, we used the model fine-tuned with the images from the forest category to perform inference on other scenes such as airport, beach, industrial, railway station, and storage tanks. Remote sensing images of the forest scene contain little texture information, which is quite different from the other scenes that we selected. In Fig. 8, the first line is the ground-truth images, and the second line displays the reconstruction results obtained by the model fine-tuned on the forest scenes. The last line is the results reconstructed by their own models. It is clear that, for all five selected scenes, the reconstruction results obtained by their own models are much better than those obtained by the model fine-tuned on the forest scenes. For the image of the railway station, the result reconstructed by the model fine-tuned on the forest scenes (the second line) tends to generate more artifacts. In addition, for the images of airport, beach, industry, and storage tanks, details such as the edges and textures contained in the images are significantly enhanced after the scene adaptation, which proves the effectiveness of the scene-adaptive strategy.

D. Validation on Other Remote Sensing Data Sets

As discussed earlier, the scene-adaptive strategy achieved satisfactory results on the AID data set. For other large-scale remote sensing images containing a mixture of various land cover scenes, we can extract the regions of interest to further improve their reconstruction results with the scene-adaptive strategy. In this section, 30 final SAMSAN models of the

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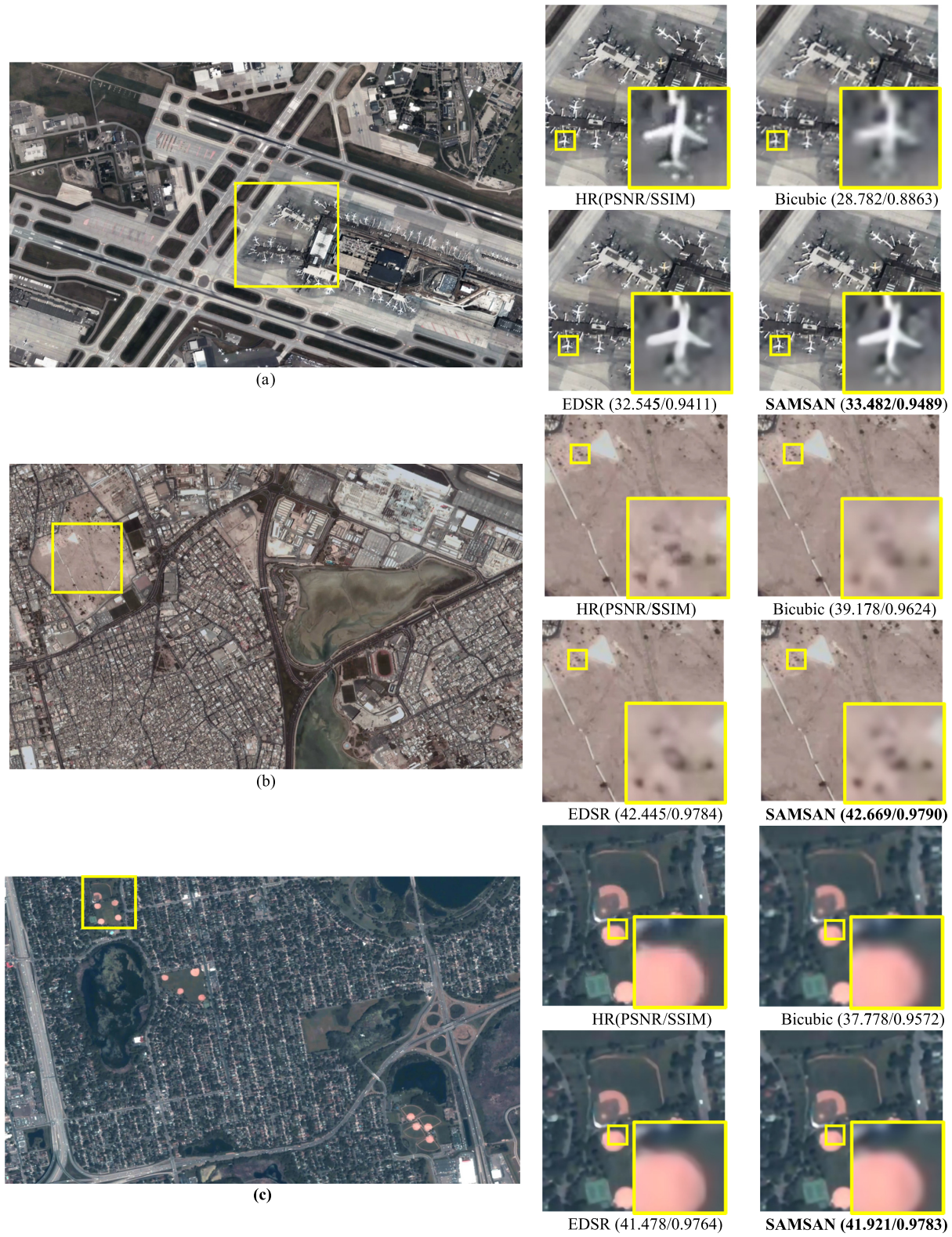


Fig. 9. Comparison results on the Jilin-1 satellite imagery for scale ×4. The figures in bold represent the best performance. (a) Airport. (b) Bare l and (c) Baseball field.

different scenes were applied to other remote sensing data sets to further evaluate this framework, including the Jilin-1 satellite imagery and the Massachusetts Road and Building Detection data set (https://www.cs.toronto.edu/~vmnih/data/).

1) Jilin-1 Imagery Data Set: The resolution of Jilin-1 imagery is 1.12 m, which is within the resolution range of the AID data set. First, we extract the region of different scenes covering the airport, bare land, and baseball-field categories from several remote sensing images of Jilin-1. We show a visual comparison of the results for the Jilin-1 satellite imagery through the different methods shown in Fig. 9. It can be clearly observed that the proposed scene-adaptive framework

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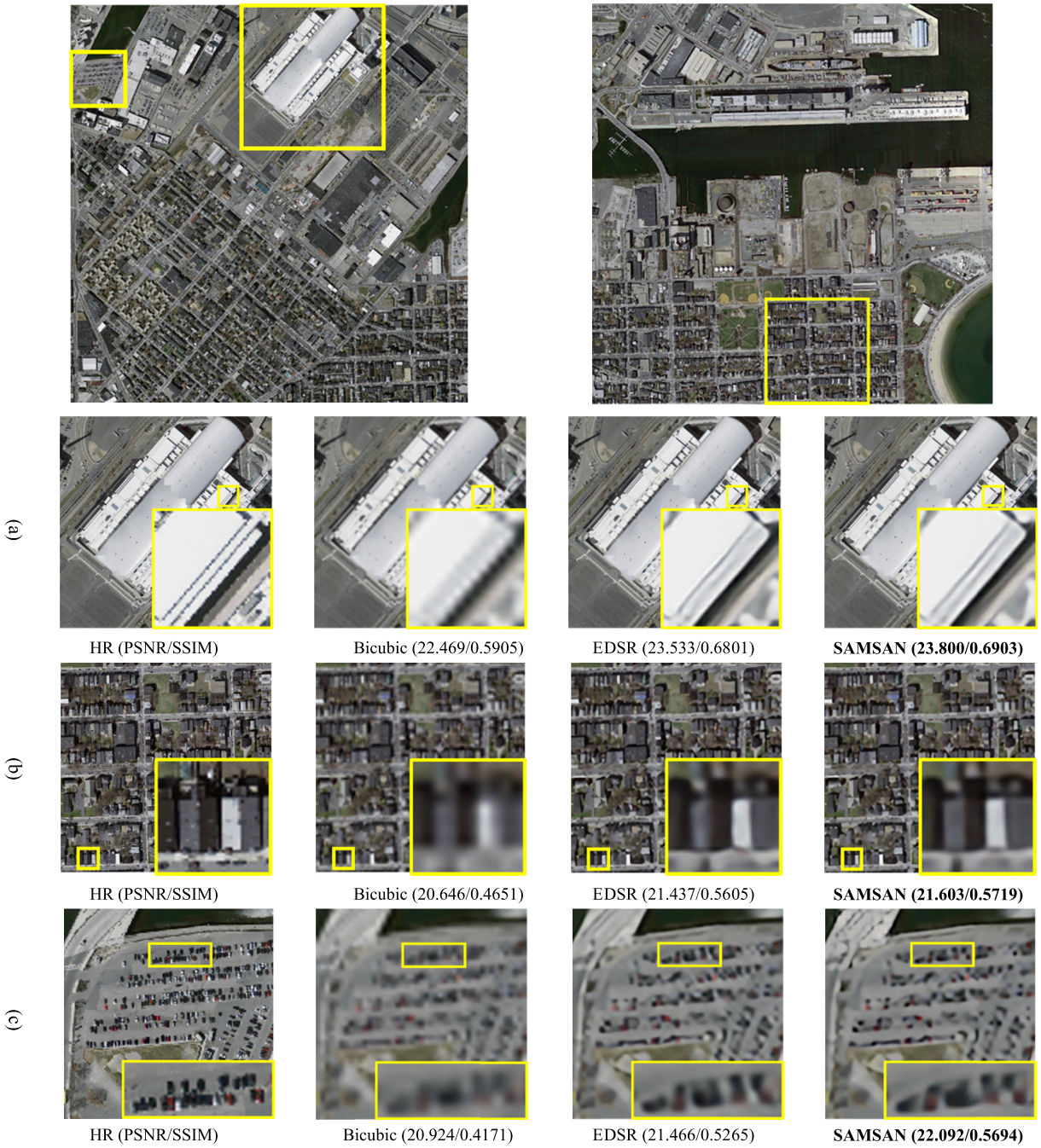


Fig. 10. Comparison results on the Massachusetts Road and Building Detection data set for scale ×4. The figures in bold represent the best performance. (a) Center. (b) Dense residential. (c) Parking.

can improve the reconstruction results for the Jilin-1 satellite imagery greatly, in both visual effect and quantitative results. The reconstructed tail and wings of the plane obtained by SAMSAN in the airport scene are the closest to that of the ground-truth image shown in Fig. 9(a). Furthermore, the PSNR value for the SAMSAN method is 0.937 dB higher than the result of EDSR. In addition, by comparing the reconstruction results for the enlarged spots in the bare land, it is obvious that the result of SAMSAN looks much clearer and is more vivid when compared with the results of the bicubic method and EDSR. For the baseball-field image, the proposed algorithm can generate sharper edges, while EDSR tends to generate more artifacts at the edges.

2) Massachusetts Road and Building Detection Data Sets: The comparison results on the Massachusetts Road and Building Detection data sets are shown in Fig. 10. Fig. 10(a)–(c) shows the reconstruction results on the scenes of the center, dense residential, and parking in the original images, respectively. The center and parking scene exist in the same original image. As shown in Fig. 10(a), the proposed MSAN model reconstructs the straight lines in the cropped image of center more accurately and obtains sharper edges than the other methods. As for the reconstruction results for the dense residential scene, the outlines of the buildings obtained by SAMSAN in Fig. 10(b) are the clearest, while the reconstructed results of EDSR appear slightly blurry. Specifically,

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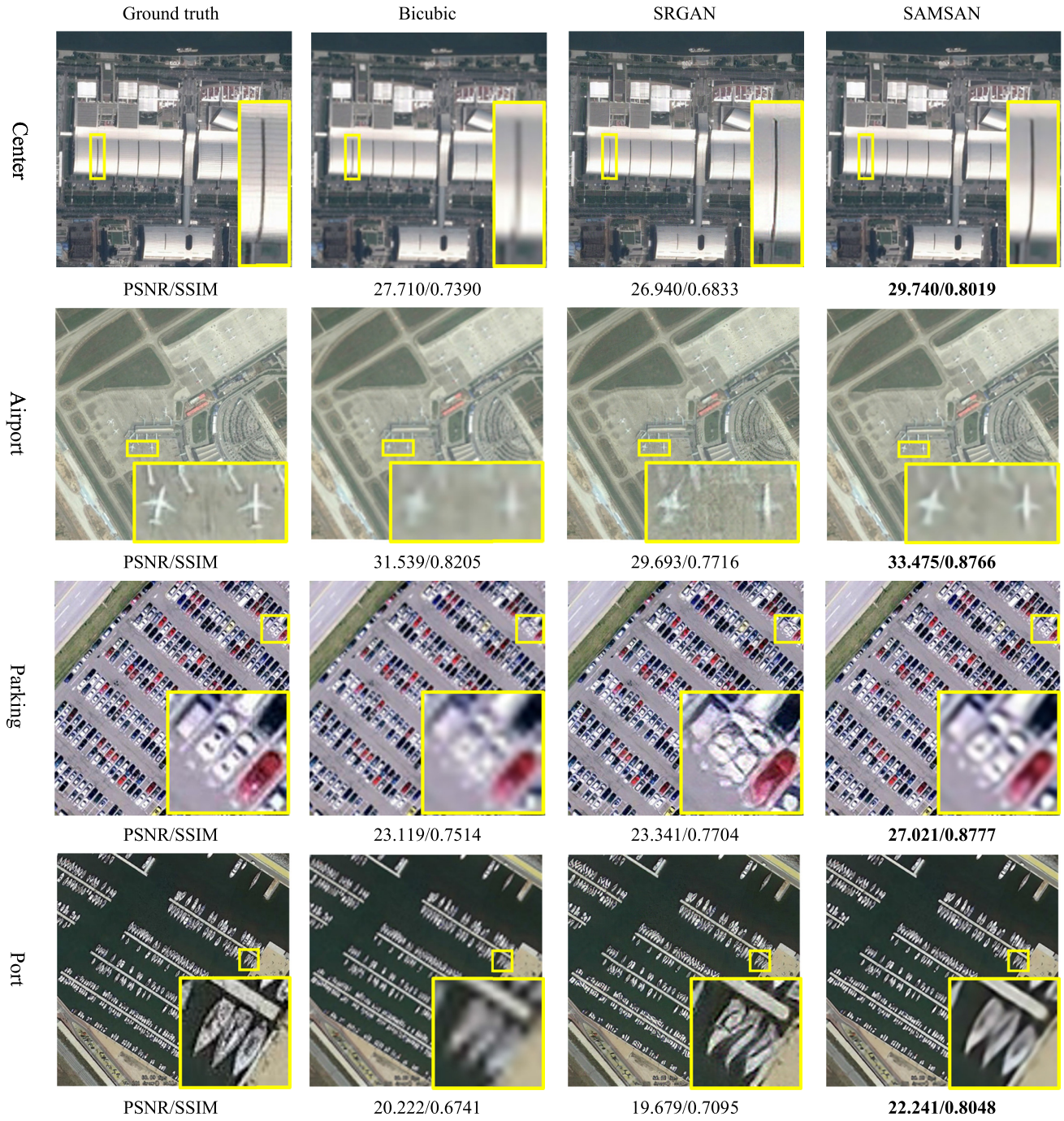


Fig. 11. Visual comparisons and PSNR/SSIM value of the proposed SAMSAN and GAN-based method on the AID data set for scale ×4.

the PSNR values of the super-resolution results of SAMSAN increase by 0.267, 0.166, and 0.626 dB, respectively, for center, dense residential, and parking. Furthermore, according to the SSIM values, it is worth noting that our proposed SAMSAN method contributes more to the SSIM than PSNR in the case of the Massachusetts data set, comparing with the results of the Jilin-1 satellite imagery. The SSIM values of the results of SAMSAN are 0.0102, 0.0114, and 0.0429 higher, respectively, for center, dense residential, and parking than those of EDSR, which means that the proposed MSAN and scene-adaptive strategy perform well in recovering the structural information of LR images.

E. Comparison With the Generative Adversarial Network (GAN)-Based Method

The preliminary experiments demonstrate the applicability of our proposed framework to various remote sensing data sets. Due to the powerful potential that the GANs shown in generating images with high perceptual quality, many GAN-based methods have been introduced into SR tasks, such as SRGAN in [35]. Also, it has been shown that GANs can provide a principled way to approach the perception-distortion bound in [51]. Therefore, in this part, we conducted some comparative experiments on the reconstruction results of the proposal and SRGAN to further investigate the effect of the

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proposed MSAN and scene-adaptive strategy. Visual examples are provided in Fig. 11, and bicubic is used as the baseline.

As shown in Fig. 11, for the objects of large scale, such as the center scene in the first line, the enlarged dark outline generated by SRGAN looks sharper, and SRGAN tends to yield a perceptually convincing result compared with that of SAMSAN. Nevertheless, when transferred to the small targets including airplanes, cars, and yachts in other scenes, it is difficult for SRGAN to recover their shape well, which is severely distorted. As we can see from the second line to the last line in Fig. 11, the textures reconstructed by SRGAN are totally in a disorder. By contrast, the small targets obtained by SAMSAN are much clearer and sharper. Consequently, benefiting from the MAFB and scene-adaptive strategy, it is obvious that SAMSAN can obtain relatively good reconstruction results for objects of various scales in remote sensing images, either large buildings or small vehicles. However, SRGAN tends to be applicable to large-scale objects only.

V. Conclusion

In this article, a multiscale channel attention network named MSAN has been proposed for remote sensing image super-resolution. To characterize the multilevel features in remote sensing images and enhance the feature expression power, we develop a MAFB by combining a multiscale activation module and a channel attention module in the RB. The proposed MSAN model outperforms the state-of-art SR algorithms of EDSR and WDSR. Moreover, in order to further utilize the underlying information among the different scene categories and characterize the feature difference between the different scenes in remote sensing images, we have presented a scene-adaptive super-resolution framework. To this end, we trained different models for images of different scenes by transfer learning and applied them to several remote sensing data sets to verify the effectiveness of the proposed framework. The experimental results obtained with the AID data set, the Jilin-1 satellite imagery, and the Massachusetts Road and Building Detection data set confirmed the effectiveness of the scene-adaptive strategy. Although the proposed algorithm achieves better results than EDSR and WDSR, the reconstruction results still lose some details, and they are dependent on accurate image classification, which may not be good enough. In the next stage, we plan to combine the deep neural network with the prior information available between the different scenes so as to fully utilize the structural information of the remote sensing images so that the super-resolution results can be further improved.

References

[1] D. Glasner, S. Bagon, and M. Irani, "Super-resolution from a single image," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 349–356.

[2] H. Su, J. Zhou, and Z.-H. Zhang, "Survey of super-resolution image reconstruction methods," *Acta Automatica Sinica*, vol. 39, no. 8, pp. 1202–1213, Jun. 2014.

[3] E. Agustsson and R. Timofte, "NTIRE 2017 challenge on single image super-resolution: Dataset and study," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1110–1121.

[4] S. Borman and R. Stevenson, "Super-resolution from image sequences–A review," in *Proc. Midwest Symp. Circuits Syst.*, Nov. 2002, pp. 374–378.

[5] L. Yue, H. Shen, J. Li, Q. Yuan, H. Zhang, and L. Zhang, "Image super-resolution: The techniques, applications, and future," *Signal Process.*, vol. 128, pp. 389–408, Nov. 2016.

[6] Q. Zhang, Q. Yuan, C. Zeng, X. Li, and Y. Wei, "Missing data reconstruction in remote sensing image with a unified spatial–temporal–spectral deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4274–4288, Aug. 2018.

[7] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial-spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[8] W. Yang, X. Zhang, Y. Tian, W. Wang, and J.-H. Xue, "Deep learning for single image super-resolution: A brief review," 2018, *arXiv:1808.03344*. [Online]. Available: https://arxiv.org/abs/1808.03344

[9] R. Keys, "Cubic convolution interpolation for digital image processing," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-29, no. 6, pp. 1153–1160, Dec. 1981.

[10] C. E. Duchon, "Lanczos filtering in one and two dimensions," *J. Appl. Meteorl.*, vol. 18, no. 8, pp. 1016–1022, Aug. 1979.

[11] S. Dai, M. Han, W. Xu, Y. Wu, Y. Gong, and A. Katsaggelos, "SoftCuts: A soft edge smoothness prior for color image super-resolution," *IEEE Trans. Image Process.*, vol. 18, no. 5, pp. 969–981, May 2009.

[12] J. Sun, Z. Xu, and H.-Y. Shum, "Image super-resolution using gradient profile prior," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2008.

[13] Q. Yan, Y. Xu, X. Yang, and T. Q. Nguyen, "Single image superresolution based on gradient profile sharpness," *IEEE Trans. Image Process.*, vol. 24, no. 10, pp. 3187–3202, Oct. 2015.

[14] A. Marquina and S. J. Osher, "Image super-resolution by TV-regularization and Bregman iteration," *J. Sci. Comput.*, vol. 37, no. 3, pp. 367–382, Dec. 2008.

[15] W. T. Freeman, T. R. Jones, and E. C. Pasztor, "Example-based super-resolution," *IEEE Comput. Graph. Appl.*, vol. 22, no. 2, pp. 56–65, Mar./Apr. 2002.

[16] S. Schulter, C. Leistner, and H. Bischof, "Fast and accurate image upscaling with super-resolution forests," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015.

[17] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[18] R. Zeyde, M. Elad, and M. Protter, "On single image scale-up using sparse-representations," in *Proc. 7th Int. Conf. Curves Surf.*, 2010, pp. 711–730.

[19] R. Timofte, V. De, and L. V. Gool, "Anchored neighborhood regression for fast example-based super-resolution," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 1920–1927.

[20] R. Timofte, V. De Smet, and L. Van Gool, "A+: Adjusted anchored neighborhood regression for fast super-resolution," in *Proc. IEEE Asian Conf. Comput. Vis.*, 2014, pp. 111–126.

[21] F. Cao, M. Cai, Y. Tan, and J. Zhao, "Image super-resolution via adaptive $l_p(0 < p < 1)$ regularization and sparse representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 7, pp. 1550–1561, Jul. 2016.

[22] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[23] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[24] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1637–1645.

[25] Y. Luo, L. Zhou, S. Wang, and Z. Wang, "Video satellite imagery super resolution via convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2398–2402, Dec. 2017.

[26] Z. Shao and J. Cai, "Remote sensing image fusion with deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1656–1669, May 2018.

[27] A. Xiao, Z. Wang, L. Wang, and Y. Ren, "Super-resolution for 'Jilin-1' satellite video imagery via a convolutional network," *Sensors*, vol. 18, no. 4, 2018, Art. no. 1194.

[28] K. Jiang, Z. Wang, P. Yi, and J. Jiang, "A progressively enhanced network for video satellite imagery superresolution," *IEEE Signal Process. Lett.*, vol. 25, no. 11, pp. 1630–1634, Nov. 2018.

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[29] J. Yu *et al.*, "Wide activation for efficient and accurate image super-resolution," 2018, *arXiv:1808.08718*. [Online]. Available: https://arxiv.org/abs/1808.08718
[30] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
[32] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017.
[33] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016.
[34] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.
[35] C. Ledig *et al.*, "Photo-realistic single image super-resolution using a generative adversarial network," 2016, *arXiv:1609.04802*. [Online]. Available: https://arxiv.org/abs/1609.04802
[36] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," 2018, *arXiv:1807.02758*. [Online]. Available: https://arxiv.org/abs/1807.02758
[37] Y. Lu, Y. Zhou, Z. Jiang, X. Guo, and Z. Yang, "Channel attention and multi-level features fusion for single image super-resolution," 2018, *arXiv:1810.06935*. [Online]. Available: https://arxiv.org/abs/1810.06935
[38] J. Gu, G. Xu, Y. Zhang, X. Sun, R. Wen, and L. Wang, "Wider channel attention network for remote sensing image super-resolution," 2018, *arXiv:1812.05329*. [Online]. Available: https://arxiv.org/abs/1812.05329
[39] J.-H. Kim, J.-H. Choi, M. Cheon, and J.-S. Lee, "RAM: Residual attention module for single image super-resolution," 2018, *arXiv:1811.12043*. [Online]. Available: https://arxiv.org/abs/1811.12043
[40] J. Li, F. Fang, K. Mei, and G. Zhang, "Multi-scale residual network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 517–532.
[41] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.
[42] G. Gao and Y. Gu, "Tensorized principal component alignment: A unified framework for multimodal high-resolution images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 46–61, Jan. 2019.
[43] Y. Gu, T. Liu, X. Jia, J. A. Benediktsson, and J. Chanussot, "Nonlinear multiple Kernel learning with multiple-structure-element extended morphological profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3235–3247, Jan. 2016.
[44] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple Kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.
[45] Y. Liu, G. Gao, and Y. Gu, "Tensor matched subspace detector for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 1967–1974, Apr. 2017.
[46] H. Song, B. Huang, Q. Liu, and K. Zhang, "Improving the spatial resolution of landsat TM/ETM+ through fusion with SPOT5," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1195–1204, Mar. 2015.
[47] W. Ma, Z. Pan, J. Guo, and B. Lei, "Achieving super-resolution remote sensing images via the wavelet transform combined with the recursive res-net," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3512–3527, Jun. 2019.
[48] T. Wang, W. Sun, H. Qi, and P. Ren, "Aerial image super resolution via wavelet multiscale convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 769–773, May 2018.
[49] S. Lei, Z. Shi, and Z. Zou, "Super-resolution for remote sensing images via local–global combined network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1243–1247, Aug. 2017.
[50] S. Hao, W. Wang, Y. Ye, E. Li, and L. Bruzzone, "A deep network architecture for super-resolution-aided hyperspectral image classification with classwise loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4650–4663, Aug. 2018.
[51] Y. Blau and T. Michaeli, "The perception-distortion tradeoff," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6228–6237.



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